

A Case Study on User Centered Design for VR Based Training in Health

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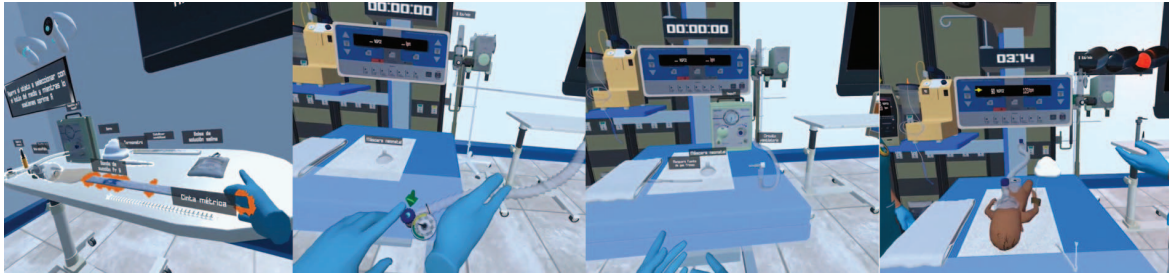


Figure 1: We present GoMi-VR, a simulator for scenarios during the Golden Minute, a newborn's first minute. GoMi-VR offers a guided introduction to both normal equipment and specialized resuscitation equipment in a delivery room. The images here show several scenarios related to (from left to right) a) Selecting items for resuscitation; b) Assembling and using the Neo-Tee; c) Assembling and using the NeoPuff; d) Performing a resuscitating of a baby with the Neo-Tee.

ABSTRACT

We present GoMi-VR, a Virtual Reality (VR) environment for training physicians in their duties during the first minute of a baby delivery. Our development process allowed us to discuss requirements remotely, create rapid prototypes, test them with experts, identify improvements, and repeat the process in a promptly manner.

Keywords: user centered design, virtual reality, training, health education

1 INTRODUCTION

This work applies established user centered design (UCD) techniques in the development of a VR-based training tool, as a way to take advantage of this design methodology for the development of experiences in this novel media. We enrich this technique by using tools that could facilitate the use of this process in a remote manner, and evaluate their use during the design and development of our training environment. In particular, we use graphs as an intermediate representation of the dialog between users and the system. We believe that this experience could show how UCD can be used in VR developments, how the development of VR-based training environments can be supported by tools, and how geographically distributed teams could collaborate in an effective manner during development.

GoMi-VR, our training environment, is an application for the Meta Quest 2 Head Mounted Display (HMD) that allows practitioners to experience several scenarios related to the first minute of a newborn's life, usually referred as the Golden Minute, including

complex scenarios that require the use of respiratory devices (Figure 1.b, c, and d.). GoMi-VR facilitates scenarios where students can learn or remember which elements should be prepared according to a case at hand (Figure 1.a). GoMi-VR also allows practitioners to experience several situations that may happen at this crucial moment, and practice the steps they have to follow in order to solve situations where a baby is not breathing. GoMi-VR represents a perfect example on how a VR-training environment can show situations that could otherwise be very difficult to reproduce.

2 RELATED WORK

As mentioned in [2], there are many challenges in the use of VR for medical education. Although we address many of the challenges they noticed during our development, such as cybersickness, attractiveness of the environment, multidisciplinary development, as well as cost minimization, the user's attitude toward the technology is of our particular interest, especially during development. UCD can improve the acceptance of novel VR technologies, through the involvement of stakeholders from early stages of development and the subsequent consideration of their concerns.

Related to development methodologies, [5] compares several design methodologies and how they could be applied to VR development. They found out user centered design to be of particular interest for VR practitioners. This theoretical result has also been found in [4] and [1]. [3] shows the development of two VR-based training environments. They also identify human-centered design as a key methodology for VR development, and they concentrate their results on how to incorporate learning theories and motivational strategies to this process. [8] describes a pipeline for the development of skill training applications, where task analysis is used as the main methodology. We use a different approach based on a dialog graph, which is commonly used in interactive, non-linear video games.

3 OUR DEVELOPMENT

GoMi-VR is a standalone application for the Oculus Quest 2 HMD that offers several experiences that may happen during the Golden Minute of a newborn. It is based on the guidelines presented at [7], which is the main reference for training physicians in several facilities.

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ties worldwide, including our Hospital. [7] comprises a book, a set of videos, a mannequin of a baby, and samples of the most common apparatus that are used for resuscitation of newborns. Despite their maturity, these materials have several shortcomings, according to our expert pediatricians: there are no changes in the skin color of the mannequin nor significant behavioral changes to the actions taken during training; most details about resuscitation are just available in the text or in videos and therefore the student's interaction is limited; just to mention a few.

GoMi-VR includes interaction with the most important devices that are used during delivery, including the Neo-Tee and the Neopuff, so that trainees can have some experience on when and how to use such devices. In total, there are twelve different cases included in GoMi-VR: Four standard procedures related to the Golden Minute without reanimation, and eight procedures that require ventilation devices. These twelve procedures are guided; i.e., there are timeouts, instructions, and suggestions on what to do at each step of the procedure. We added four more cases without guidance, from the standard procedures for the Golden Minute, that could be useful for evaluation. In this last four cases we expect students to perform all activities in one minute or less, as it is expected in the Golden Minute procedure.

4 RESULTS AND DISCUSSION

We performed early testing at Hospital Militar Central with experts in our system, physicians, and medical students of at least fourth year. All subjects had previous knowledge about the Golden Minute. At the beginning, discussions were performed in scenarios in Mozilla Hubs or in AltVR, as described in [6]. During development we produced 34 versions, 17 of which were released in the last three months of development. Most versions were tested by two or three people, since their purpose was very focused. Almost two months before the end of our project we performed a larger test with 13 subjects, in order to test all functionality and the overall behavior of the entire experience.

Subjects were asked to rate how close to reality is the experience in this simulator from 1 to 7 where 1 meant "not close at all" and 7 "faithful to reality". Results from all versions are shown in Figure 2.

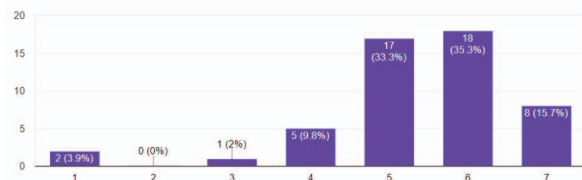


Figure 2: Distribution of answers to the question "How close is the experience to the real scenario?"

We also asked how much subjects knew about these procedures before and after the experience, and results are shown in Figure 3. Although future studies should be done in order to better measure learning and retention, it is encouraging to see that subjects of all versions feel that they improve their knowledge after using the simulator.

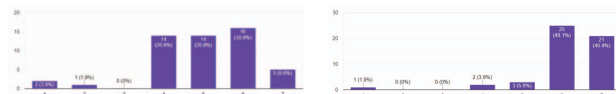


Figure 3: Reported knowledge before (left) and after (right) the experience. The results show more concentration on the higher levels of learning after the experience.

Task load measurements were high, both mentally and physically, which is related to the high pace of the tests we asked users to perform in just 20 minutes. Hopefully, in a more normal learning scenario in the future and with proper training we could facilitate the use more. The results related to presence were mostly favorable, especially in the latter versions. However, initial findings were key in order to improve each version. For example, issues related to cybersickness were reported from the beginning, we did improvements in each new version, and the overall experience can still be improved in future versions.

Cybersickness was an issue in the first versions and even though it has improved, it can be further improved. Initially, we left subjects the option to move either by joystick or by teleportation. Movement by joystick is more familiar to subjects with previous experience in videogames, so many subjects preferred such option. However, this causes more discomfort in VR. Later versions use only teleportation, which significantly reduces this effect, but it is less familiar. This requires now extra training, but it is worth the time in order to avoid discomfort. Also, it seems women tend to suffer more from cybersickness than men in our experience, but more studies are necessary to be conclusive.

Since our environment can be used at any time, subjects can train at will whenever they have the time, and since possible mistakes are not public, they feel more relaxed during the experience. Due to the variety of cases implemented and their length, it is impossible to see everything in one interaction of 20 minutes, and we implemented ways to *jump* to a particular part of the scenario. This was very useful for testing of particular issues, and for review of particular content, but we still have to define proper ways to include this functionality into the experience. However, subjects were willing to try GoMi-VR in order to see all the cases available.

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